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XRA-GS: X-ray Attenuation-Aligned Gaussian Splatting for Sparse Tomographic View Synthesis

Abstract

3D Gaussian Splatting has been extended to X-ray novel view synthesis, yet existing methods retain the gradient-driven densification rules from natural-image 3DGS without modification. Under sparse CT acquisition, these rules cause Gaussians to accumulate near high-contrast anatomical boundaries while interior tissues that determine X-ray attenuation remain under-represented. We present ***XRA-GS****, an X-ray Attenuation-Aligned Gaussian Splatting framework that corrects this structural imbalance by redesigning the Gaussian evolution pipeline. XRA-GS introduces three components: a support-profile seeding strategy that initializes Gaussians from the coarse attenuation support estimated by FDK reconstruction; a geometry-aware pruning step that removes Gaussians concentrated near anatomical boundaries; and an adaptive density modulation step that applies position-dependent density refinement throughout training. Experiments on five organs under a sparse-view CT protocol demonstrate that XRA-GS achieves state-of-the-art novel view synthesis quality against six representative baselines from both visible-light sparse-view synthesis and X-ray CT methods.*

1. Introduction

Computed tomography (CT) is an essential imaging technique for noninvasively examining the internal structure of an object, and is widely used in clinical diagnosis, image-guided intervention, and industrial inspection [FDK84; AK84; WYD08]. During a CT acquisition, an X-ray source rotates around the object while a detector records multi-angle attenuation measurements, from which cross-sectional images or projections at unseen angles can be recovered through tomographic reconstruction. Reducing the number of acquired views is highly desirable, since it directly lowers radiation dose, shortens scan time, and broadens applicability to patients who cannot sustain long, stable acquisitions [LMC*23; WHL*24]. In this work, we study *Sparse Tomographic View Synthesis*: from only a handful of X-ray projections, the model must predict the line-integrated attenuation observed at unseen acquisition angles, so that each synthesized projection remains consistent with the Beer–Lambert imaging geometry rather than merely visually similar to the inputs. A reliable solution supplies auxiliary views for anatomical localization, lesion-range inspection, and intraoperative planning under severely limited sampling, where conventional analytical reconstruction breaks down [WN19].

Visible-light and X-ray imaging are governed by fundamentally different rendering equations, and this difference dictates where representation capacity should reside. In visible-light 3DGS, a pixel is formed by front-to-back α-compositing of an emission–absorption volume rendering integral [KKLD23; Max95]: the accumulated transmittance decays multiplicatively along the ray, so for opaque scenes the integrand is dominated by Gaussians near the first-hit surface while contributions from deeper Gaussians are progressively masked. The surface-clustered Gaussian layout that emerges under α-compositing is therefore a direct consequence of this transmittance-induced concentration, and is consistent with the surface-aligned geometry that surface-oriented 3DGS variants exploit explicitly [GL24; HYC*24]. In X-ray imaging, however, the detector reading follows the Beer–Lambert law [KS01]: after taking the logarithm, the recorded quantity becomes a linear, order-independent line integral of the attenuation coefficient along the ray, with no transmittance decay and no occlusion. Consequently, every Gaussian on the path contributes on equal footing, and capacity must be distributed *along the entire ray path* rather than collapsed onto any single layer. Existing X-ray Gaussian methods such as X-Gaussian, R2-Gaussian, X-Field, and DGR have correctly replaced α-compositing with a radiative line-integral renderer [CLW*24; ZLC*24; WTW*25; WLG*25], and sparse-view 3DGS pipelines such as FSGS, CoR-GS, and DNGaussian further compensate for missing observations through depth or co-regularization priors [ZFJW24; ZLY*24; LZB*24]. Yet all of them inherit a Gaussian evolution rule that was originally designed for surface-centric α-compositing [KKLD23; RNSZ24].

Crucially, the issue is not gradient-driven densification itself. Splitting or cloning Gaussians where the projection-error gradient is large is a sound idea, and our method retains exactly this mechanism. The issue is that under a line-integral renderer, the residual gradient is still concentrated at high-attenuation-contrast interfaces, such as bone–soft-tissue boundaries, which almost coincide with the high-color-contrast interfaces that drive visible-light densification. A naïve port therefore keeps spawning Gaussians at anatomical boundaries even though the rendering equation now asks for capacity *along* the path. Under sparse views, this bias surfaces not as insufficient coverage but as a structural *capacity misallocation*: a substantial fraction of Gaussians is spent on boundary-driven redundancy, while interior path segments that determine the attenuation integral remain under-represented. The top row of Fig. 1 shows the natural fit between α-compositing and surface-clustered Gaussians; the bottom row shows that when the same evolution rule is reused under line-integral rendering, capacity still collapses to boundaries and leaves the deep path empty. Fixing this requires *augmenting*, not discarding, gradient-driven densification with attenuation-aligned decisions about where to seed, what to prune, and how to refine.

Motivated by this analysis, we propose **XRA-GS**, an X-ray Attenuation-Aligned Gaussian Splatting framework for Sparse Tomographic View Synthesis. Rather than appending auxiliary branches to a conventional densification backbone, **XRA-GS** keeps the gradient-driven backbone of standard 3DGS and adds three attenuation-aligned stages that act on *where* Gaussians are seeded, retained, and refined, as illustrated in Fig. 1. Support-Profile Seeding (SPS) anchors initialization to a coarse FDK attenuation support [FDK84], so that Gaussians populate physically meaningful X-ray paths instead of starting from a contrast-biased prior; Geometry-aware Pruning (GAP) recycles Gaussians that accumulate near high-contrast anatomical interfaces during training, draining boundary-driven redundancy at its source; and Adaptive Density Modulation (ADM) applies position-dependent online refinement so that retained Gaussians carry interior attenuation rather than uniformly inflated density. Across diverse organs and view budgets, this attenuation-aligned evolution reallocates representation capacity from boundary redundancy to attenuation-relevant interior structures.

Our main contributions are summarized as follows:

- We identify *capacity misallocation* as the failure mode when sparse-view X-ray Gaussian methods inherit a visible-light evolution rule under a line-integral renderer: gradient-driven densification keeps clustering Gaussians at attenuation boundaries while interior path segments that determine the line integral remain starved of capacity.
- We propose **XRA-GS**, an X-ray Attenuation-Aligned Gaussian Splatting framework that augments standard gradient-driven densification with three coordinated attenuation-aligned stages—Support-Profile Seeding (SPS) for path-anchored initialization from coarse FDK support, Geometry-aware Pruning (GAP) for boundary-redundancy recovery, and Adaptive Density Modulation (ADM) for position-dependent online refinement—without modifying the X-ray renderer.
- Extensive experiments on multi-organ sparse-view CT show that **XRA-GS** delivers consistent gains in projection fidelity over state-of-the-art X-ray Gaussian and sparse-view 3DGS baselines, while reducing the number of Gaussians and training cost; ablations confirm that SPS, GAP, and ADM each contribute complementary capacity reallocation.

2. Related Work

2.1. Sparse-view Novel View Synthesis

3D Gaussian Splatting (3DGS) represents a scene with anisotropic Gaussians and renders through differentiable rasterization [KKLD23]. Under sparse-view novel view synthesis, per-scene optimization methods diverge in how they explain failure: FSGS attributes degradation to insufficient coverage and augments densification with proximity-guided unpooling and depth priors [ZFJW24]; CoR-GS treats the problem as unstable optimization and introduces dual-field co-regularization with co-pruning [ZLY*24]; DNGaussian injects geometric priors through depth normalization [LZB*24]. A parallel feed-forward line moves geometry completion into large-scale pretraining [CLTS24; CXZ*24; XPW*25; LFY*25]. Explicit-and-hybrid representations such as K-Planes [FYT*23] demonstrate that plane-decomposed spatial encodings provide continuous positional context at low cost; we adopt this encoding in ADM for position-dependent density refinement. Regardless of the direction, these methods share an error/gradient-driven densification backbone that prioritizes high-contrast surfaces. In CT, where each detector pixel integrates attenuation along the entire traversed path rather than sampling surface radiance, this evolutionary logic systematically misallocates capacity toward anatomical

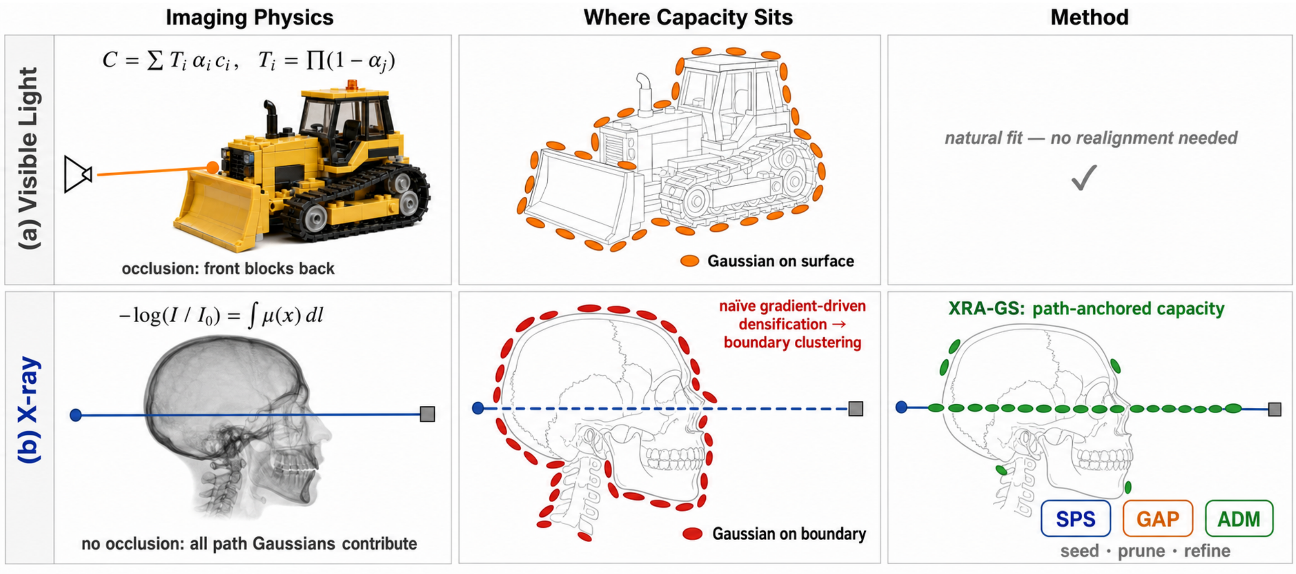

Figure 1: ***Visible-light vs. X-ray Gaussian rendering and the capacity gap they impose.*** *(a) Visible-light α-compositing concentrates capacity on visible surfaces, where front Gaussians occlude back ones. (b) X-ray line integrals weight every Gaussian along the ray equally, so naïve gradient-driven densification still clusters at attenuation boundaries;* ***XRA-GS*** *realigns the evolution rule through SPS, GAP, and ADM to anchor capacity along the path.*

boundaries and underrepresents the interior structures that also participate in attenuation.

2.2. X-ray Novel View Synthesis and Tomographic Reconstruction

Learning-based sparse-view X-ray/CT methods fall into two families. Implicit methods extend neural attenuation fields with X-ray imaging structure: SAX-NeRF jointly models line dependencies and projection context [CWY*24], while Geometry-Aware Attenuation Learning shifts toward population-level encoder-decoder inference [LFL*25].

The explicit line redesigns Gaussians around X-ray attenuation. X-Gaussian introduced radiative Gaussians and differentiable X-ray rasterization [CLW*24]; R2-Gaussian further aligned the representation with reconstruction consistency through integration-bias correction and differentiable voxelization [ZLC*24]. Subsequent work strengthens physical consistency: X-Field introduces material-aware ellipsoids and path segmentation for physics-aligned attenuation modeling [WTW*25]—its contribution targets the imaging equation rather than the evolution rule governing where Gaussians accumulate, leaving capacity misallocation from boundary-driven densification as an unaddressed bottleneck in sparse-view CT; DGR targets discretized voxel reconstruction [WLG*25]; GR-Gaussian addresses artifact suppression through graph-structured gradients [YMS*25]; X^2-Gaussian extends radiative Gaussians to 4D dynamic CT [YCZ*25]; and Layer-Based CT partitions volumes into anatomical depth layers for slice-wise reconstruction [Kua*25]. These extensions target 4D dynamic CT, discretized voxel recovery, artifact-suppression graphs, or slice-wise volumetric reconstruction—objectives that fall outside the sparse-view novel-view projection protocol our experiments evaluate, and we therefore omit them from the main comparison. Our work belongs to the explicit radiative-Gaussian line, but its contribution is not another physical branch. Instead, we argue that even with a physically aligned renderer, sparse-view CT still suffers if Gaussian capacity allocation inherits the boundary-chasing logic of visible-light densification. This misallocation operates at three distinct stages: *initialization* that ignores the X-ray attenuation support, *boundary over-densification* during training, and *undifferentiated point-wise density updates* in later refinement. SPS, GAP, and ADM each address one of these stages in turn, forming an attenuation-aligned Gaussian evolution pipeline rather than a modified rendering branch.

3. Method

Figure 1 already summarizes the core distinction at the pipeline level: **XRA-GS** does not keep stacking auxiliary branches on top of an existing densification backbone, but instead reorders how Gaussians are initialized, reclaimed, and refined. We now turn to the implementation level and show how this attenuation-aligned evolution pipeline is instantiated by SPS, GAP, and ADM on top of the radiative Gaussian backbone of R2-Gaussian.

3.1. Problem Formulation

Let the sparse-view observation set be $\mathcal{P} = \{(I_v, \pi_v)\}_{v=1}^{V}$, where I_v denotes the v-th X-ray projection and π_v denotes its corresponding acquisition geometry. Given a coarse FDK reconstruction volume V_{FDK} obtained from these observations, our goal is to learn an explicit radiative Gaussian representation in a per-case optimization setting so that unseen-angle target projections can be synthesized from only a few observed projections. Although this task is naturally coupled with volumetric density recovery, our primary optimization target remains novel-view quality in the projection domain.

3.2. 3DGS and X-ray Rendering Background

To clarify that our problem is not simply "adding another prior," we first contrast the pixel formation mechanisms of visible-light 3DGS and X-ray projection. In standard visible-light 3DGS, the color of a pixel is typically accumulated through front-to-back compositing of multiple splats ordered along the viewing ray:

$$\mathbf{C}(\mathbf{r}) = \sum_{i=1}^{N_{\mathbf{r}}} T_i \, \alpha_i \, \mathbf{c}_i, \qquad T_i = \prod_{j<i} (1 - \alpha_j), \tag{1}$$

where $\mathbf{r}$ is the camera ray, and α_i and $\mathbf{c}_i$ denote the opacity and radiance attributes of the i-th splat. The core intuition is that a pixel is formed by the accumulation of local surface contributions.

In X-ray imaging, however, a detector pixel corresponds more closely to the remaining energy of a single ray after traversing internal materials. Its physical measurement can be written as

$$\begin{aligned} I(\mathbf{r}) &= I_0 \exp\left(- \int_{\mathbf{r}} \mu(\mathbf{x}) \, \mathrm{d}l \right), \\ -\log \frac{I(\mathbf{r})}{I_0} &= \int_{\mathbf{r}} \mu(\mathbf{x}) \, \mathrm{d}l, \end{aligned} \tag{2}$$

where $\mu(\mathbf{x})$ is the attenuation coefficient at spatial location $\mathbf{x}$ and $\mathrm{d}l$ is the differential path length along the ray. In other words, an X-ray pixel is not a sum of surface colors; it is the integral of attenuation contributions from different materials along a single path. What matters is how much of each material occupies the path, rather than how strong the local response is at one boundary point.

We represent the scene as a Gaussian set $\mathcal{G} =$

$\{g_i\}_{i=1}^{N}$, where each Gaussian is written as

$$g_i = (\mathbf{p}_i, \Sigma_i, \rho_i, \mathbf{c}_i), \tag{3}$$

denoting the center, covariance, density, and appearance/radiative parameters, respectively. For any view π_v, the R2-Gaussian-style radiative renderer can be written as

$$\hat{I}_v = \mathcal{R}(\mathcal{G}; \pi_v). \tag{4}$$

Except for our new SPS, GAP, and ADM modules, the radiative rasterization, densification, and differentiable voxelization settings follow R2-Gaussian [ZLC*24]. Importantly, X-Gaussian and R2-Gaussian have already aligned the renderer with X-ray radiative projection. The mismatch we target lies mainly in the Gaussian evolution rule rather than in the imaging equation itself. In this baseline, Gaussians are still usually duplicated or split according to projection-error gradients. This is effective for improving surface coverage in natural scenes, but in CT it tends to keep pushing the evolving Gaussian set toward high-gradient anatomical boundaries. The model therefore uses the right X-ray renderer, yet still organizes Gaussians with a visible-light-style growth logic, which is precisely the issue our three modules address.

3.3. Overall Framework

As shown in Figure 2, **XRA-GS** instantiates the attenuation-aligned Gaussian evolution pipeline as three sequential modules embedded into the explicit radiative Gaussian backbone of R2-Gaussian. SPS seeds Gaussians into more relevant anatomical regions during initialization using the coarse FDK reconstruction. GAP then periodically reclaims locally redundant Gaussians after densification, preventing high-gradient boundaries from dominating Gaussian growth. Finally, ADM refines the density distribution of retained Gaussians based on position-aware context during later optimization. In short, we do not redesign X-ray physics; we reorganize the optimization loop itself into path anchoring, boundary de-redundancy, and local refinement. Algorithm 1 summarizes the complete training procedure.

Algorithm 1 XRA-GS Training Pipeline

Require: Sparse projections $\mathcal{P}$, FDK volume V_{FDK}, iterations T
Ensure: Optimized Gaussian set $\mathcal{G}^*$
Stage 1 — Initialization (SPS):
  Compute density-weighted sampling $p(\mathbf{x})$ from V_{FDK} // *Eq.* (5)
  Mix with uniform: $q(\mathbf{x}) \leftarrow \alpha/|\Omega| + (1-\alpha)\, p(\mathbf{x})$ // *Eq.* (6)
  Sample M centers; set scales and densities from V_{FDK}
Stage 2 — Main optimization loop:
for $t = 1, \ldots, T$ **do**
  Render $\hat{I}_v \leftarrow \mathcal{R}(\mathcal{G}; \pi_v)$ for observed views // *Eq.* (4)
  Compute loss $\mathcal{L}$ // *Eq.* (16)
  Update $\mathcal{G}$, ADM parameters via back-propagation
  if $t \bmod T_{\text{densify}} = 0$ **then**
    Standard densification (clone / split)
    if $t \in [T_{\text{gap}}, T_{\text{gap,end}}]$ **then**
      GAP: compute proximity s_i; prune by mask m_i // *Eqs.* (7)–(8); $T_{gap} = 2K$, $T_{gap,end} = 20K$
    end if
  end if
  if $t \geq T_{\text{adm}}$ **then**
    ADM: predict $\tilde{o}(\mathbf{x})$; modulate ρ_{final} // *Eq.* (15)
  end if
end for
return $\mathcal{G}^* \leftarrow \mathcal{G}$

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Figure 2: ***Overall framework of XRA-GS.*** *Given sparse X-ray projections* $\{(I_v, \pi_v)\}_{v=1}^{V}$ *and a coarse FDK reconstruction* V_{FDK}*, SPS (purple) performs path-anchored initialization along the attenuation support outside the optimization loop. An iterative training loop then interleaves ADM (orange, position-aware density modulation) and GAP (teal, gradient-aware redundancy recovery), supervised end-to-end by the rendering loss. Solid arrows denote data flow; dashed arrows denote gradient flow. The output is a novel-view X-ray projection together with the recovered tomographic volume.*

3.4. Support-Profile Seeding (SPS)

SPS initializes Gaussians in anatomically relevant regions using support-profile cues from the coarse FDK reconstruction. Initialization strongly influences the downstream optimization trajectory, yet neither of the standard 3DGS initializers fits CT well: uniform sampling cannot distinguish the anatomical importance of air, soft tissue, and bone, while SfM-based point-cloud initialization is unstable under gray-value overlap and weak local texture in X-ray projection [CLW*24]. Crucially, we do not use a vague spatial prior. Instead, we draw on three concrete signals contained in the FDK reconstruction: foreground anatomical support, low-frequency attenuation layout, and a coarse intensity hierarchy that differentiates air, soft tissue, and bony structures. The sparse-view CT literature also widely treats FDK/FBP reconstruction as a coarse CT volume with streak artifacts whose main value lies in global structure layout and useful frequency content rather than reliable fine detail [WCS*22; LMC*23; LFZ*25; ZLC*24]. Based on these structural properties of the FDK reconstruction, we use the coarse density volume V_{FDK} produced by FDK and define a density-weighted sampling distribution over a discrete voxel grid Ω:

$$p(\mathbf{x}) = \frac{(\max(V_{\mathrm{FDK}}(\mathbf{x}), \epsilon))^{\gamma}}{\sum_{\mathbf{x}' \in \Omega} (\max(V_{\mathrm{FDK}}(\mathbf{x}'), \epsilon))^{\gamma}}, \tag{5}$$

where ϵ avoids zero probability in background regions and γ controls sampling concentration.

Density weighting alone would over-concentrate Gaussians near the brightest bony structures and could mistake the strongest sparse-view FDK responses for precise geometry. We therefore mix uniform sampling with the support-profile distribution:

$$q(\mathbf{x}) = \alpha \frac{1}{|\Omega|} + (1 - \alpha) p(\mathbf{x}), \quad \mathbf{x} \in \Omega, \tag{6}$$

and independently sample M initial centers $\{\mathbf{x}_m\}_{m=1}^{M}$ from $q(\mathbf{x})$, where M is the number of initial Gaus-

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Figure 3: ***Support-Profile Seeding (SPS).*** *(a) From the sparse projections, a coarse FDK volume is reconstructed; the horizontal attenuation profile along an anatomical scan line exposes the support-profile cue (purple curve). (b) SPS converts the attenuation magnitude into a density-weighted sampling map (viridis) and mixes it with a uniform prior, so capacity concentrates on anatomically relevant support while a small uniform term retains global coverage. (c) Compared with random initialization (gray dots), SPS produces an initial Gaussian set (purple dots) already aligned with the attenuation support before any optimization step.*

sians. This design preserves global coverage while allocating more Gaussians to anatomical regions with meaningful attenuation support, without collapsing entirely onto dense bony boundaries. After sampling the locations, we initialize Gaussian densities from the FDK values and set scale parameters according to the local sampling density, yielding 50K initial Gaussians. In all experiments, α and γ are selected by a small local search and then fixed to avoid scene-specific manual tuning.

3.5. Geometry-aware Pruning (GAP)

GAP periodically removes locally redundant Gaussians based on spatial proximity and gradient activity. In CT, gradient-driven densification often keeps duplicating Gaussians around bone-soft tissue boundaries because these regions produce larger projection-error gradients. However, high gradient does not always imply missing coverage; it may also indicate that the same anatomical boundary is being represented repeatedly by neighboring Gaussians. Instead of following the FSGS logic of adding support points in sparse regions [ZFJW24], GAP explicitly measures local crowding in world coordinates and uses it for redundancy recovery in over-densified regions.

Specifically, for each Gaussian G_i, let $\mathcal{N}_K(i)$ denote the index set of its K nearest neighbors excluding itself. We compute the mean Euclidean distance to these neighbors as a proximity score:

$$s_i = \frac{1}{K} \sum_{j \in \mathcal{N}_K(i)} \|\mathbf{p}_i - \mathbf{p}_j\|_2. \tag{7}$$

When s_i falls below a threshold τ, the Gaussian lies in an overly crowded region. To avoid deleting Gaussians that are still learning useful structure, we further introduce a gradient-activity filter. Let $\bar{g}_i$ denote the accumulated L_1 statistic of the positional gradient of this Gaussian over the current densification cycle. We define a retention mask $m_i \in \{0, 1\}$ as

$$m_i = \begin{cases} 1, & s_i \geq \tau \ \vee \ \bar{g}_i \geq \delta, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

where $m_i = 0$ means that the Gaussian is treated as a redundancy candidate in the current cycle. In implementation, GAP does not replace the original 3DGS densification; it is a periodic recovery step applied afterwards. We first generate candidate Gaussians through the standard duplication/splitting rule and then use m_i to filter redundant ones. In each cycle, we prune at most a fraction β_{prune} of the total Gaussians, and the module is triggered periodically from 2K to 20K iterations. To avoid suddenly making local support too sparse after pruning, we only

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Figure 4: *Geometry-aware Pruning (GAP).* *(a) After standard gradient-driven densification, Gaussians accumulate around high-attenuation boundaries, producing visible over-densification clusters. (b) For each Gaussian, GAP measures local crowding by the mean Euclidean distance to its K nearest neighbors and combines it with the accumulated gradient activity over the current densification cycle: a Gaussian is flagged as a redundancy candidate (red $\times$) only if both KNN distance is below τ and gradient activity is below δ; otherwise it is retained (green $\circ$). (c) After removing at most a fraction β_{prune} of the redundant candidates, the cleaned distribution preserves the attenuation support while freeing capacity for under-served interior regions.*

shrink the covariance of newly retained Gaussians:

$$\Sigma_i^{\text{new}} = \eta \Sigma_i, \quad 0 < \eta < 1. \tag{9}$$

GAP therefore regularizes the densification process by suppressing locally redundant Gaussians, preventing high-gradient boundaries from continuously monopolizing representation capacity.

3.6. Adaptive Density Modulation (ADM)

ADM predicts position-dependent density offsets to refine the density distribution of retained Gaussians. Even after better initialization and redundancy control, density updates remain highly heterogeneous across anatomy: bone usually corresponds to higher attenuation, air regions in the lung should remain low-density, and soft tissue often requires smoother transitions. If all Gaussians share the same point-wise gradient update rule, the model struggles to express this spatial heterogeneity. We therefore encode Gaussian centers with K-Planes and use a lightweight decoder that predicts only density-related offsets and confidence gates, without rewriting Gaussian centers, covariances, or other attributes. The role of ADM is local density refinement rather than serving as the sole source of performance.

It is important to emphasize that, although ADM contains an MLP, it is not an offline pre-trained prediction head and does not rely on extra density supervision. Both the K-Planes features and the dual-head MLP parameters are updated only during optimization on the current case, and they are trained end-to-end through the main rendering loss. In this sense, ADM is an online joint-optimization branch that adaptively amplifies or suppresses density responses in different regions while the main radiative Gaussian backbone is being trained.

Given a position $\mathbf{x}$, we first project it onto normalized 2D coordinates $\Pi_{uv}(\mathbf{x})$ on three orthogonal planes and then bilinearly sample the corresponding feature maps:

$$f_{uv}(\mathbf{x}) = \mathcal{B}(P_{uv}, \Pi_{uv}(\mathbf{x})), \quad uv \in \{xy, xz, yz\}, \tag{10}$$

$$F(\mathbf{x}) = \operatorname{concat}\big(f_{xy}(\mathbf{x}), f_{xz}(\mathbf{x}), f_{yz}(\mathbf{x})\big), \tag{11}$$

where $\mathcal{B}(\cdot)$ denotes bilinear sampling, P_{xy}, P_{xz}, and P_{yz} are the three orthogonal feature planes, and $\operatorname{concat}(\cdot)$ denotes channel-wise concatenation. K-Planes encoding was originally introduced for explicit spatiotemporal radiance fields and provides continuous spatial context at low cost [FYT*23]. Based on this encoding, a dual-head MLP predicts a density offset $\Delta\sigma(\mathbf{x})$ and a confidence gate $g(\mathbf{x})$:

$$\big(\Delta\sigma(\mathbf{x}), g(\mathbf{x})\big) = h(F(\mathbf{x})). \tag{12}$$

We then center the offsets by the batch mean and apply a view-dependent scaling:

$$\tilde{o}(\mathbf{x}) = \beta(t)\, s_{\text{view}}\, r_{\max}\, g(\mathbf{x}) \cdot \big(\Delta\sigma(\mathbf{x}) - \overline{\Delta\sigma}\big), \tag{13}$$

where $\overline{\Delta\sigma}$ is the mean density offset of the current batch, $r_{\max}$ is the maximum modulation amplitude, and s_{view} is a view-count scaling factor (concretely, $s_{\text{view}} = 0.5$, 0.7, 1.0 for 2, 3, and 4 input views, respectively). To prevent modulation from disrupting basic geometric convergence early in training, we use a warm-up/hold/decay schedule for $\beta(t)$:

$$\beta(t) = \begin{cases} \frac{t}{t_w}, & 0 \le t < t_w, \\ 1, & t_w \le t < t_h, \\ \exp\big(-\frac{t-t_h}{\kappa}\big), & t \ge t_h. \end{cases} \tag{14}$$

where t_w marks the end of the warm-up phase, t_h marks the end of the hold phase, and κ is the exponential decay time constant. The final density is

$$\rho_{\text{final}}(\mathbf{x}) = \rho_{\text{base}}(\mathbf{x}) \cdot \big(1 + \tilde{o}(\mathbf{x})\big). \tag{15}$$

Therefore, ADM learns not a global density boost, but a spatial-context-aware relative correction that better matches the locally continuous yet regionally heterogeneous structure of CT data.

3.7. Training Loss

The three modules of **XRA-GS** and the radiative rendering backbone of R2-Gaussian form a single end-to-end training pipeline. During training, we supervise reprojection results on the observed views with the following overall loss:

$$\mathcal{L} = \mathcal{L}_1 + \lambda_{\text{dssim}}\mathcal{L}_{\text{dssim}} + \lambda_{\text{3DTV}}\mathcal{L}_{\text{3DTV}} + \lambda_{\text{pTV}}\mathcal{L}_{\text{pTV}}, \tag{16}$$

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Figure 5: ***Adaptive Density Modulation (ADM).*** *(a) Three orthogonal K-Planes feature maps (viridis) encode position-dependent spatial context for every Gaussian center. (b) The concatenated K-Planes feature is decoded by a lightweight dual-head MLP that predicts a density offset* $\Delta\sigma(\mathbf{x})$ *(orange) and a confidence gate* $g(\mathbf{x}) \in [0,1]$ *(slate); the modulated density combines them through* $\rho_{final} = \rho_{base} \cdot (1+\tilde{o})$*. (c) The resulting density difference map (inferno) shows that ADM amplifies density in regions where the attenuation support requires fine detail and damps modulation where evidence is uncertain.*

where

$$\mathcal{L}_1 = \frac{1}{V}\sum_{v=1}^{V} \|I_v - \hat{I}_v\|_1, \tag{17}$$

$$\mathcal{L}_{\text{dssim}} = \frac{1}{V}\sum_{v=1}^{V} \text{DSSIM}(I_v, \hat{I}_v), \tag{18}$$

$$\mathcal{L}_{\text{3DTV}} = \sum_{i=1}^{N} \sum_{d\in\{x,y,z\}} |\nabla_d \rho_i|\,, \tag{19}$$

$$\mathcal{L}_{\text{pTV}} = \sum_{p\in\{xy,xz,yz\}} \omega_p(t)\,\text{TV}(P_p). \tag{20}$$

Here, $\omega_p(t)$ is the time-varying regularization weight for the p-th feature plane; it follows the same warm-up/hold/decay schedule as $\beta(t)$, ramping linearly from 0 to ω_0 over the first t_w iterations, holding at ω_0 until t_h, then decaying exponentially with time constant κ. $\mathcal{L}_1$ and $\mathcal{L}_{\text{dssim}}$ jointly constrain projection reconstruction quality, $\mathcal{L}_{\text{3DTV}}$ encourages spatial smoothness in volumetric density, and $\mathcal{L}_{\text{pTV}}$ imposes planar total-variation regularization on the K-Planes features. During training, the plane features and MLP parameters of ADM are jointly updated with the backbone Gaussian parameters under the same loss, so density modulation is learned online rather than fixed in advance. Since SPS only changes initialization, GAP is triggered periodically during training, and ADM applies lightweight density modulation, the basic X-ray rendering process at inference remains unchanged and the extra cost is concentrated in training.

4. Experiments

4.1. Experimental Setup

Data and task setting. We evaluate on the five-organ dataset from the X-Gaussian repository [CLW*24]: Chest (LIDC-IDRI) and Head, Abdomen, Foot, and Pancreas (open scientific visualization dataset). Following the NAF/X-Gaussian preprocessing and projection protocol, we construct sparse-view settings by retaining only 2, 3, or 4 input projections per scene. Models are trained exclusively on observed projections and evaluated on held-out novel angles.

Baselines. We compare against six methods in two groups. *Direct X-ray/CT baselines:* X-Gaussian [CLW*24] (radiative Gaussian splatting), R2-Gaussian [ZLC*24] (reconstruction-consistency regularization), and X-Field [WTW*25] (material-aware implicit modeling) represent the state of the art for X-ray novel view synthesis. *Natural-scene sparse-view 3DGS references:* CoR-GS [ZLY*24], DNGaussian [LZB*24], and FSGS [ZFJW24] characterize how gradient-driven densification degrades under tomographic projection. X-Field is re-evaluated under our identical protocol and denoted as X-Field* in the result tables and figures.

Metrics. Following the official X-Gaussian evaluation script, PSNR2D and SSIM2D are computed on $[0, 1]$-normalized projections averaged over all held-out views of each scene and then over the five organs. PSNR2D is $20\log_{10}(1/\sqrt{\text{MSE}})$; SSIM2D uses an 11×11 Gaussian window ($\sigma = 1.5$, $C_1 = 0.01^2$, $C_2 = 0.03^2$).

Implementation details. SPS uses $\alpha = 0.2$, $\gamma = 1.0$, and 50K initial Gaussians. GAP uses $K = 5$ neighbors, threshold $\tau = 0.015$, maximum pruning ratio $\beta_{\text{prune}} = 0.02$, and gradient threshold $\delta = 0.0002$. ADM uses K-Planes at resolution 64×32 and activates after 15K iterations. All models are trained for 30K iterations.

4.2. Main Results

Table 1 reports PSNR2D across all five organs and three sparsity levels. **XRA-GS** achieves the best five-organ average at every sparsity level. Gains over R2-Gaussian are $+0.11/+0.39/+0.07$ dB under 2/3/4-view settings; gains over X-Field* are $+0.87/+3.25/+2.79$ dB. The most pronounced margin occurs at 3 views, consistent with our central argument: capacity misallocation is most directly addressable when projection evidence is sufficient to guide redistribution yet too limited for standard densification to converge to a spatially consistent solution.

Table 2 reports SSIM2D averages for the three methods with accessible logs under our unified protocol. **XRA-GS** leads at 2- and 3-view settings and ties R2-Gaussian at the third decimal place in 4-view, confirming the same trend as PSNR2D.

Figure 6 provides qualitative comparisons across all five organs under the 2-view setting. The natural-scene sparse-view baselines (CoR-GS, DNGaussian,

Table 1: *Main PSNR2D comparison under the unified five-organ, 2/3/4-view protocol. Blank entries denote same-protocol results that are unavailable in the source logs, so we do not estimate or extrapolate them. Best and second-best results are highlighted in bold and underline, respectively.*

Method	Chest			Head			Abdomen		
	2v	3v	4v	2v	3v	4v	2v	3v	4v
DNGaussian	14.30	16.02	15.76	15.72	18.04	16.24	19.33	23.51	24.72
CoR-GS	14.96	18.55	20.29	16.24	20.36	23.40	18.50	22.45	23.03
FSGS	17.86	19.99	21.03	19.43	21.53	24.91	18.75	23.34	23.63
X-Gaussian	17.79	20.29	20.90	19.62	21.52	24.46	18.93	23.54	23.85
R2-Gaussian	<u>21.05</u>	<u>26.22</u>	<u>25.47</u>	<u>23.54</u>	<u>26.60</u>	**28.66**	<u>24.85</u>	<u>29.26</u>	<u>30.56</u>
X-Field*									
XRA-GS	**21.26**	**26.81**	**25.83**	**23.63**	**26.63**	<u>28.52</u>	**24.97**	**29.73**	**30.89**

Method	Foot			Pancreas			Avg		
	2v	3v	4v	2v	3v	4v	2v	3v	4v
DNGaussian	20.17	23.51	19.33	16.15	22.43	25.47	17.13	20.70	20.30
CoR-GS	20.17	25.53	26.83	16.15	23.25	23.96	17.20	22.03	23.50
FSGS	<u>22.17</u>	25.37	27.08	<u>23.03</u>	25.27	26.22	20.25	23.10	24.57
X-Gaussian	**22.34**	25.48	26.92	**23.21**	25.12	25.55	20.38	23.19	24.34
R2-Gaussian	19.39	<u>28.48</u>	**30.04**	17.83	<u>28.61</u>	**30.90**	<u>21.33</u>	<u>27.83</u>	<u>29.13</u>
X-Field*							20.57	24.97	26.41
XRA-GS	19.55	**28.49**	<u>29.86</u>	17.80	**29.43**	**30.90**	**21.44**	**28.22**	**29.20**

FSGS) produce characteristic boundary leakage and streak artifacts under tomographic projection; the direct X-ray baselines (X-Gaussian, R2-Gaussian, X-Field*) preserve global attenuation structure more faithfully. **XRA-GS** further reduces background diffusion and local over-accumulation near high-contrast anatomical boundaries, consistent with the quantitative gains in Table 1.

Figure 7 provides two complementary analyses. The upper portion enlarges the Chest and Foot regions marked in Figure 6, isolating boundary and structure recovery at the local scale. The lower portion evaluates a Chest 2-view case at three novel angles (0°, 45°, 90°), confirming that the advantage of **XRA-GS** over R2-Gaussian is stable across viewing directions rather than concentrated at a single angle.

Same-protocol comparison with X-Field. The X-Field* row in Table 1 confirms that the gains of

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5 organs (rows) × 8 cols (GT / CoR-GS / DNGaussian / FSGS / X-Field* / X-Gaussian / R2-Gaussian/ **XRA-GS**) + error map col, 2-view setting

Figure 6: ***Qualitative comparison on five organs, 2-view setting.*** *Rows: Chest, Head, Abdomen, Foot, Pancreas (top to bottom). Columns: GT, CoR-GS, DNGaussian, FSGS, X-Field*, X-Gaussian, R2-Gaussian, and* ***XRA-GS*** *(Ours, last column). Natural-scene references produce streak artifacts and boundary leakage; direct X-ray baselines preserve global attenuation structure;* ***XRA-GS*** *further suppresses local over-accumulation and background diffusion. Rightmost column: error maps (inferno, shared vmax). Red boxes mark regions enlarged in Figure 7.*

Table 2: *SSIM2D comparison (five-organ average). Per-organ SSIM2D logs for CoR-GS, DNGaussian, FSGS, and X-Gaussian are unavailable under our unified 2/3/4-view protocol; we therefore report same-protocol averages only for the three methods with accessible logs.* ***Bold****: best; underline: second-best.*

Method	2v Avg.	3v Avg.	4v Avg.
X-Field*	0.717	0.850	0.880
R2-Gaussian [ZLC*24]	0.794	0.903	**0.924**
XRA-GS (Ours)	**0.797**	**0.904**	**0.924**

XRA-GS are not an artifact of protocol asymmetry. Re-evaluated under our identical 2/3/4-view setting, X-Field* trails **XRA-GS** by a consistent average margin across all sparsity levels.

4.3. Module and Initialization Analysis

Table 3 reports a progressive ablation starting from the R2-Gaussian backbone. SPS raises the 2-view average PSNR2D from 21.27 to 21.44 dB, establishing that path-anchored initialization provides a meaningful head start in the most under-determined regime. GAP delivers the largest PSNR2D increment, improving the 3-view average from 28.01 to 28.22 dB by reclaiming capacity from over-densified boundary regions. The final **XRA-GS** row matches or improves the available PSNR2D measurements across all view counts, while intermediate SSIM2D values are pending from the experiment logs.

Figure 8 translates the progressive ablation into local visual evidence. Residual maps $|\hat{I} - I_{\mathrm{gt}}|$ are shown below each prediction. This visual analysis is used to inspect where the progressive modules change the reconstruction, while the corresponding

Placeholder: `assets/fig/experiment_fig_qual_detail_consistency.png`

Upper: zoom-in Chest + Foot (4 cols: GT / R2-Gaussian/ X-Gaussian / **XRA-GS**)

Lower: Chest 2-view at 0°/45°/90° (3 rows × 4 cols)

Figure 7: ***Local detail and cross-view consistency.*** Upper half: *Zoom-ins from the red boxes in Figure 6 for Chest (left) and Foot (right), comparing GT, R2-Gaussian, X-Gaussian, and **XRA-GS**. **XRA-GS** reduces streak spread and boundary diffusion in both regions.* Lower half: *A Chest 2-view case rendered at* 0°, 45°, *and* 90°. *The quality advantage of **XRA-GS** persists across all three angles, indicating consistent global attenuation reconstruction rather than angle-specific improvement.*

Table 3: *Progressive component ablation (five-organ average). Modules are added incrementally on top of the R2-Gaussian backbone. GAP delivers the largest available PSNR2D increment, while SPS benefits the most under-determined 2-view regime. Intermediate SSIM2D values are pending from experiment logs, so the current table supports PSNR2D trends only for intermediate configurations.* ***Bold:*** *best per column.*

Configuration	PSNR2D↑ (dB)			SSIM2D↑		
	2v	3v	4v	2v	3v	4v
Baseline (R2-Gaussian)	21.27	27.80	29.10	–	–	–
+ SPS	21.44	28.01	29.16	–	–	–
+ SPS + GAP	21.44	28.22	29.20	–	–	–
XRA-GS (SPS + GAP + ADM)	**21.44**	**28.22**	**29.20**	**0.797**	**0.904**	**0.924**

SSIM2D for intermediate configurations is pending from experiment logs.

intermediate SSIM2D values remain pending from the experiment logs.

Figure 9 is reserved for visualizing the cumulative effect of the three modules on Gaussian spatial distribution once the corresponding Gaussian-center and count logs are available. Until those logs are confirmed, the mechanism is evaluated through the available quantitative ablation in Table 3.

Table 4 (left) isolates the initialization strategy. SPS-guided seeding outperforms random initialization at all sparsity levels; the full **XRA-GS** further improves upon SPS-only through GAP and ADM, confirming that path-anchored initialization and capacity recovery are complementary operations.

4.4. View-count Analysis

Table 4 (right) reports the view-count trend for four representative methods. **XRA-GS** achieves the highest five-organ average PSNR2D at every sparsity level. The relative gain over R2-Gaussian is largest at 3 views, where capacity misallocation is most directly observable; at 4 views the reconstruction-consistency regularization in R2-Gaussian partially closes the

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Chest 3-view — top: GT / Baseline / +SPS / +SPS+GAP / Full **XRA-GS**

bottom: residual maps $|\hat{I} - I_{\mathrm{gt}}|$ (inferno); ΔSSIM2D annotated under each col

Figure 8: *Progressive module ablation, Chest 3-view.* *Top row: rendered projections for GT, Baseline (R2-Gaussian), +SPS, +SPS+GAP, and full **XRA-GS**. Bottom row: residual maps $|\hat{I} - I_{gt}|$ (inferno, shared vmax with Figure 6). Quantitative intermediate SSIM2D values are pending from experiment logs and are therefore not used for module-level claims here.*

Placeholder: `assets/fig/experiment_fig_spatial_distribution.png`

4 panels: Uniform Init (≈50K) → +SPS (≈42K) → +SPS+GAP (≈38K) → Full **XRA-GS** (≈35K); attenuation profile strip above

Figure 9: *Gaussian spatial distribution evolution.* *Four panels (left to right): uniform initialization, after SPS, after GAP pruning, and full **XRA-GS**, for a representative Chest organ. Top strip: FDK attenuation profile (magma shading). The final figure will use logged Gaussian centers and counts rather than schematic estimates.*

gap, consistent with the applicability boundary discussed in Section 4.7.

4.5. Computational Cost

Table 5 is reserved for training and inference cost under the 3-view setting. The current manuscript keeps the table structure in place, but Time, final Gaussian count, GPU memory, and FPS must be filled from experiment logs before making quantitative efficiency claims.

4.6. Hyperparameter Sensitivity

Figure 10 is reserved for local sweeps of four key hyperparameters in the Chest 3-view setting: the SPS mixture coefficient α, the GAP proximity threshold τ, the GAP pruning ratio β_{prune}, and the ADM acti-

Table 4: ***Left: Initialization strategy ablation*** *(five-organ average PSNR2D, dB). SPS-guided seeding outperforms random initialization at all sparsity levels; the full* ***XRA-GS*** *further gains from GAP and ADM, confirming that path-anchored initialization and capacity recovery are complementary.* ***Right: View-count trend*** *for four representative methods (five-organ average PSNR2D, dB).* ***XRA-GS*** *leads at every sparsity level; the relative advantage over R2-Gaussian is largest at 3 views, where capacity misallocation is most directly observable.* ***Bold****: best; underline: second-best.*

Initialization	**2v**	**3v**	**4v**
Random (no SPS)	21.27	27.80	29.10
SPS-guided (no GAP, ADM)	<u>21.44</u>	<u>28.01</u>	<u>29.16</u>
XRA-GS (full)	**21.44**	**28.22**	**29.20**

Five-organ avg PSNR2D↑ (dB). Per-organ and SSIM2D data in supplementary.

Method	**2v**	**3v**	**4v**
FSGS [ZFJW24]	20.25	23.10	24.57
X-Gaussian [CLW*24]	20.38	23.19	24.34
R2-Gaussian [ZLC*24]	<u>21.33</u>	<u>27.83</u>	<u>29.13</u>
XRA-GS (Ours)	**21.44**	**28.22**	**29.20**

Five-organ avg PSNR2D↑ (dB). CoR-GS and DNGaussian omitted for brevity (full data in Table 1).

Table 5: *Computational cost comparison under the 3-view setting (five-organ average). Train Time: end-to-end training minutes. #Gaussians: final primitive count ($\times 10^3$) after densification/pruning. GPU Mem: peak memory during training (GB). FPS: rendering frames per second at inference. PSNR2D: five-organ average (dB), from Table 1. Cost columns will be filled from matched experiment logs before quantitative efficiency claims are made.* ***Bold****: best per column.*

Method	**Time↓ (min)**	**#Gauss.↓ (K)**	**Mem.↓ (GB)**	**FPS↑**	**PSNR2D↑ (dB)**
X-Gaussian [CLW*24]	–	–	–	–	23.19
R2-Gaussian [ZLC*24]	–	–	–	–	27.83
XRA-GS (Ours)	–	–	–	–	**28.22**

"–" indicates values pending from experiment logs requested in `assets/ask/experiment_2026-05-23_T01_T02_T04_main_metrics_efficiency.md`. PSNR2D is taken from Table 1.

vation iteration. The default configuration ($\alpha = 0.2$, $\tau = 0.015$, $\beta_{\text{prune}} = 2\%$, ADM activation at 15K) will be assessed once the sweep metrics are available from the experiment logs.

Placeholder: `assets/fig/experiment_fig_hparam.png`

2×2 panels: (a) SPS α; (b) GAP τ; (c) GAP β_{prune}; (d) ADM activation

Primary y-axis: SSIM2D (solid Crimson); secondary: PSNR2D (dashed); • = default

Figure 10: ***Hyperparameter sensitivity, Chest 3-view.*** *Each panel sweeps one hyperparameter with all others fixed at the default (marked •). Solid Crimson: SSIM2D↑; dashed: PSNR2D↑. (a) SPS mixture coeff.* $\alpha \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$*; (b) GAP threshold* τ*; (c) GAP pruning ratio* $\beta_{prune} \in \{0, 1, 2, 5, 10\}\%$*; (d) ADM activation iteration (fraction of 30K iters). The final sweep figure will be populated from experiment logs.*

4.7. Discussion and Limitations

The experiments confirm that in the moderately sparse regime, the primary bottleneck is capacity allocation rather than Gaussian count. Gains are most pronounced at 3 views, where sufficient projection evidence exists for path-anchored initialization and boundary de-redundancy to take effect, yet too little evidence for standard densification to converge to a spatially consistent solution.

Two boundaries constrain the current method. At 2 views the FDK reconstruction used by SPS is insufficiently reliable, so the initialization advantage partially erodes. At 4 views the reconstruction-consistency regularization in R2-Gaussian is strong enough to compensate for boundary-chasing effects, reducing the relative advantage of **XRA-GS**. Figure 11 illustrates both failure modes: in Foot 2-view and Pancreas 2-view, fine-grained high-frequency detail remains unrecoverable when projection evidence is fundamentally insufficient, regardless of initialization or capacity management.

Several directions remain open. First, SPS relies on FDK quality, which degrades sharply below two views; learned or uncertainty-aware initializers could relax this constraint. Second, the GAP threshold and pruning ratio are fixed organ-agnostically; organ-adaptive scheduling may improve per-organ performance. Third, all experiments assume parallel-beam geometry; extension to cone-beam and dynamic CT acquisition is left for future work. Additional per-organ SSIM2D breakdowns and single-module ablations will be included only after the corresponding same-protocol logs are confirmed.

5. Conclusion

XRA-GS demonstrates that visible-light densification rules impose a systematic capacity-allocation failure in transmissive X-ray imaging that persists even after the rendering equation is physically corrected. By intervening at three stages of Gaussian evolution—path-anchored initialization (SPS), boundary redundancy recovery (GAP), and position-dependent local refinement (ADM)—the framework redirects representation capacity from over-densified anatomical boundaries toward the interior attenuation-relevant structures that conventional gradient-driven growth neglects. Across a five-organ benchmark with 2/3/4 input views against six representative baselines, **XRA-GS** achieves the best average PSNR2D at every sparsity level, with SSIM2D reported for methods whose same-protocol logs are available. The strongest PSNR2D improvement appears in the moderately sparse 3-view regime, where sufficient projection evidence allows capacity reallocation to take effect.

Placeholder: `assets/fig/experiment_fig_failure_cases.png`

2 rows: Foot 2-view (top) / Pancreas 2-view (bottom)

4 cols: GT / R2-Gaussian/ **XRA-GS** / error map (inferno); failure desc. inset right ($\leq$25 words)

Figure 11: ***Representative failure cases, 2-view setting.*** *Rows: Foot (top), Pancreas (bottom). Columns: GT, R2-Gaussian,* ***XRA-GS****, and* ***XRA-GS*** *error map (inferno, shared vmax with Figure 6). Zoom-in boxes mark the primary failure region.* ***XRA-GS*** *reduces overall error spread relative to R2-Gaussian but cannot resolve fine structural detail when projection evidence is fundamentally absent. Failure descriptions ($\leq$25 words) are inset at the right of each row.*

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